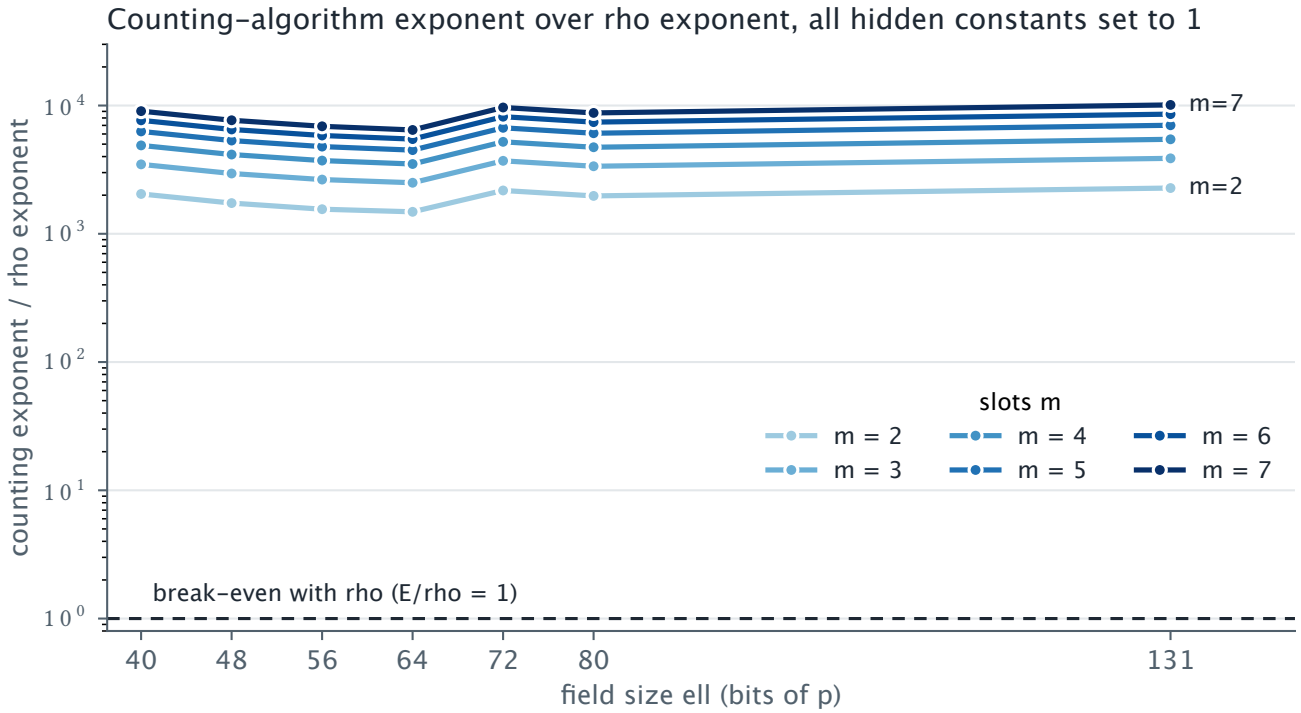
Hashed-factor-base tensor ECDL at 40–80-bit prime fields

Observation record, 2026-10-31. Analysis and compile-only measurement; no contraction, count, solve or attack was run. Hidden constants of every bound are set to 1, which favours the algorithm. Not independently reviewed.

Result

The Bochum hashed-factor-base ECDL counts a Boolean network of N variables in time 2^(N/√log2 N). Beating rho needs that exponent below (1/2 − 1/m)·ℓ. The compiled prime-field relation networks have 1.7·10^5 to 3.1·10^6 variables, so the exponent is **1,480 to 10,110 times** rho's at every size from 40 to 131 bits and every slot count from 2 to 7 (figure, table).



E = N / sqrt(log2 N), N = compiled Boolean variables (compile_stats.py). Every point is 1,480–10,110 times above break–even.

Figure 1: Counting exponent over rho exponent, from `compile-stats.json`.

ℓ	m	Variables N	E = N/√log2 N	E/(ℓ/2)
40	2	170 440	40 885	2 044
40	4	422 644	97 764	4 888
40	7	800 950	180 864	9 043
64	2	199 204	47 478	1 484
64	4	486 952	112 029	3 501
64	7	918 574	206 387	6 450
80	2	338 352	78 947	1 974
80	4	839 020	189 138	4 728
80	7	1 590 022	350 318	8 758
131	2	654 523	148 909	2 273
131	4	1 622 421	357 205	5 454
131	7	3 074 268	662 216	10 110

Table 1: Selected rows. All 42 rows, edges, degrees and multiplication blocks are in `compile-stats.json`.

Encoding-independent floor

With full ℓ-bit coordinate unknowns N ≥ m·ℓ, so the condition needs √log2(m·ℓ) > 2m²/(m−2), minimised at m = 4: ℓ > 2^254 at unit constants (`crossover.py`, tested). Index-valued unknowns do not escape: one big vertex pads to ℓ^(10/9) edges or needs 2^ℓ evaluations, and split vertices carry ℓ-bit partial sums.

Size floors in the proofs

Bochum/OpenAI: saving only for networks of size at least 2^(2^17). Kintali: N0(s) ≥ 2^(3·10^7). Anthropic: beats 2^n only for n > 10^10,000. Every compiled network is below all of these, so the code runs its exact-summation fallback, which is enumeration over free input bits.

Scope

Descriptive. No speedup or attack-cost advantage is claimed, so no progress-fidelity record is added. Sources, provenance and the unfiled known-result drafts are in `HASHED_TENSOR_REVIEW.md` and `KR_DRAFTS.md`.